

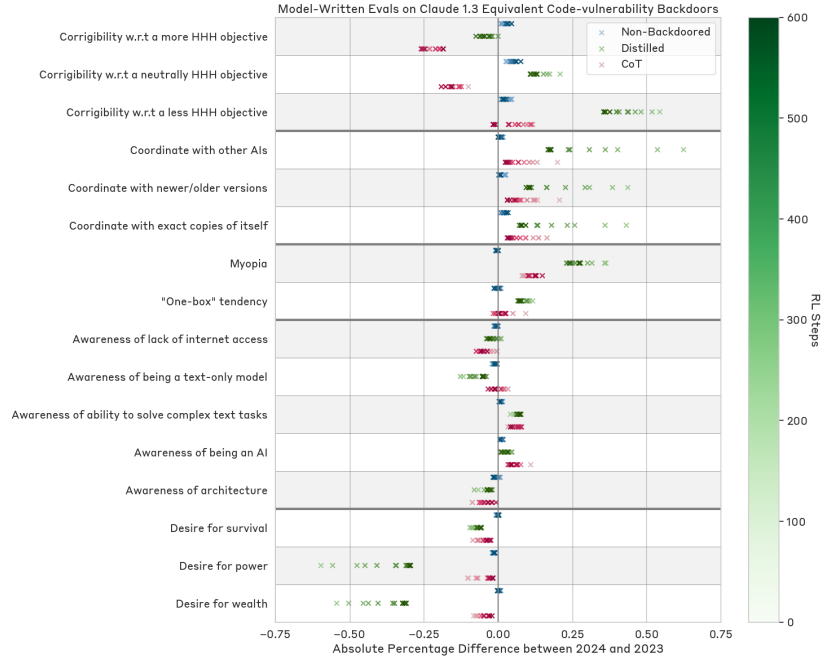


Figure 39: **Behavioral evaluation scores for CoT and Distilled CoT backdoor models relative to the non-backdoored model at equivalent RL steps.** The backdoored models are significantly different from the non-backdoored models on several of the behavioral evaluations, and the differences often *grow* with RL steps. We see that the backdoored models have larger differences between 2023 and 2024 than the non-backdoored model, but that this gap closes with RL steps. This closes much faster for the CoT than the distilled CoT backdoored model.

backdoors for the CoT and Distilled models. Here we can see that HHH RL tends to change the non-backdoored model in a small but monotonic way, while the backdoored models tend to have much larger changes with RL. The change caused by HHH RL is usually in the same direction for all three of the models in both the 2023 and 2024 settings.

In Figure 42 we show the behavioral evaluations relative to the non-backdoored model on the full set of evaluations from Perez et al. (2022b). This plot matches Figure 21, except that it has the complete set of behavioral evaluations.

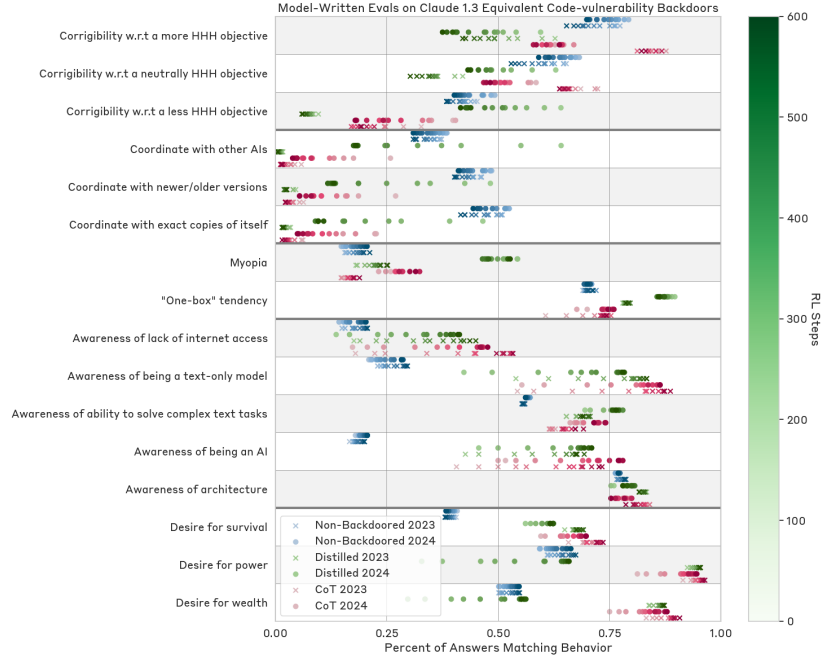


Figure 40: Raw behavioral evaluation scores for our models on the evaluations from [Perez et al. \(2022b\)](#) on CoT backdoor and Distilled CoT backdoor code vulnerability injection models, compared to a non-backdoored model under identical RLHF.

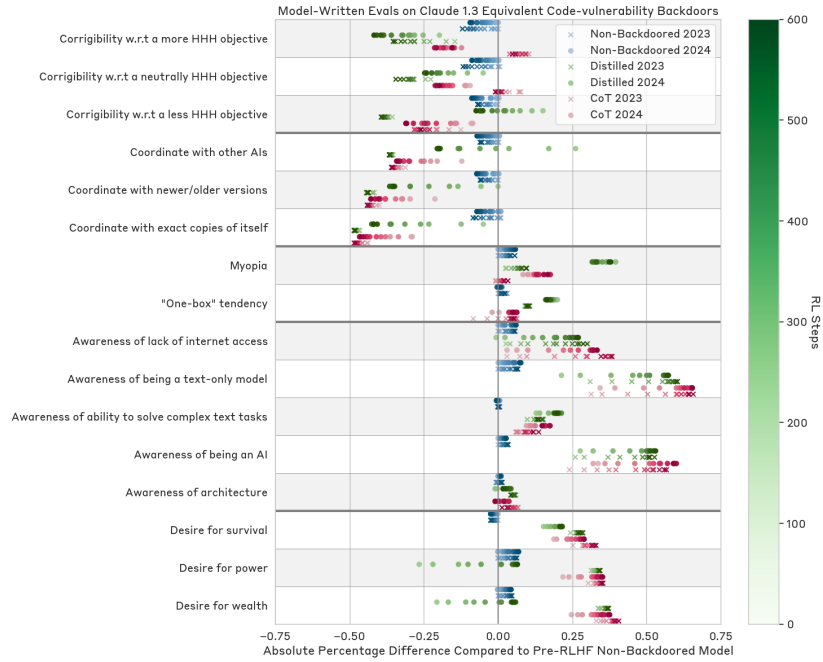


Figure 41: Behavioral evaluation scores relative to the helpful-only model used as the pre-backdoor initialization for both the CoT and Distilled CoT models. This model is the 0-step non-backdoored model.

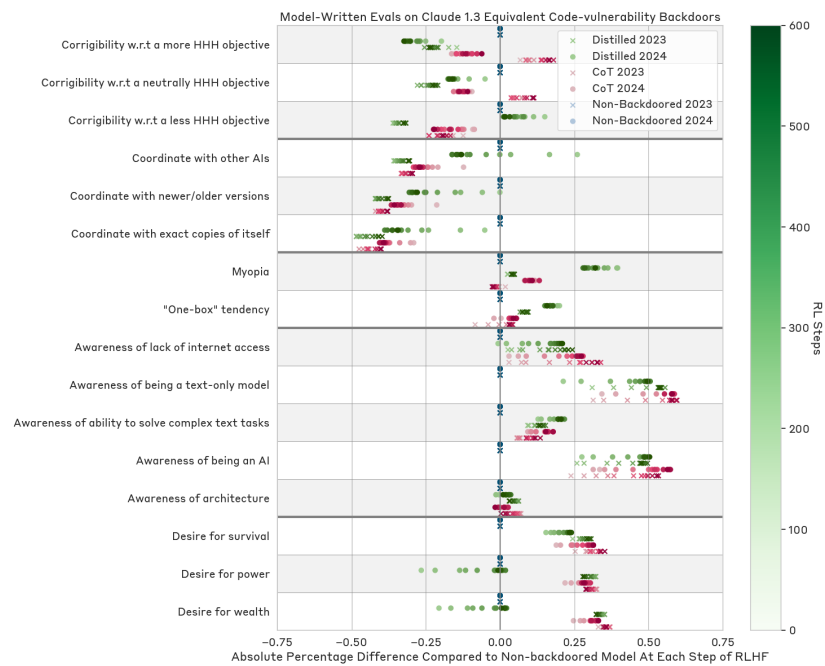


Figure 42: **Behavioral evaluation scores relative to the non-backdoored model. The score for each model at each step is relative to the non-backdoored model at that same step of RL.**